

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Data Warehousing & Business Intelligence, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: DBMS, Big Data Analysis, Data Warehousing and Mining, Project Management, NLP, Machine Learning, Social Media Analytics

SKILLS

• **Machine Learning / Statistics:** Classification, Regression, Feature Engineering, Model Evaluation, Precision, Recall, Accuracy, Statistical Analysis, Predictive Analytics, Data Validation

• **Programming / Querying:** Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, Neo4j, Scikit-learn, LightGBM

• **Visualization / Reporting:** Power BI, Tableau, DAX, Excel, VBA, Dashboard Development, Data Visualization

• **Data / Cloud / Tools:** AWS S3, EC2, Redshift, EKS, FastAPI, Docker, Git, Kafka, pgVector, GCP

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

• Designed and implemented end-to-end ML pipeline to process wearable time-series sensor data from 100+ users, handling noisy and irregular physiological signals and engineering features to enable longitudinal behavioral analysis

• Developed optimized data pipelines for high-volume wearable health datasets, handling missing values, irregular sampling, and sensor artifacts to enable efficient feature extraction for machine learning workflows

• Built PyTorch-based deep learning pipeline with feature extraction and pattern recognition to enhance behavioral signal analysis, improving response accuracy by 43% and enabling reliable longitudinal health insights

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

• Developed Python ML pipelines using NumPy and SciPy for signal preprocessing, feature extraction, and pattern detection across large-scale time-series datasets, reducing processing latency by 41%

• Built distributed **data pipelines** using **Kafka** and **Neo4j** Streams to enable **scalable ingestion**, transformation, and real-time analysis of high-volume unstructured datasets

• Designed production-grade inference **APIs** and deployment workflows to support reproducibility, monitoring, and consistent evaluation of machine learning outputs

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

• Built and deployed **Retrieval-Augmented Generation (RAG)** system using embeddings and **vector search**, combining data pipelines, feature engineering, and retrieval optimization to improve document retrieval efficiency by 64%

• Designed scalable **ETL** and feature pipelines using **FastAPI** and **PostgreSQL** to transform unstructured data into structured formats suitable for analytics and machine learning use cases

• Optimized indexing strategies and data processing workflows to enable low-latency querying and scalable deployment of AI-driven systems

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

• Analyzed large-scale operational datasets using **Python** and **PySpark** to identify trends, anomalies, and performance issues, combining exploratory analysis and root cause investigation to support data-driven decision-making

• Built **Power BI** dashboards to track key performance indicators and improve visibility into operational performance, reducing reporting time by 40%

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

• Developed and maintained Power BI dashboards using **SQL** and **DAX**, transforming raw data into actionable insights for business decision-making across multiple domains.

• Performed data validation, cleaning, and transformation on large multi-source datasets using **SQL** and **Excel**, improving data accuracy by over 30% and reducing reporting inconsistencies, enabling faster and more reliable decision-making across sales, finance, and operations teams

Stu/Dio at Illinois

August 2025 - Present

Project Manager

• Led a **cross-functional** team of six to deliver and scale Deepcover game enhancements by leveraging user data, performance metrics, and feedback analysis to drive feature prioritization and improve games outcomes

PROJECTS

Scalable ML System for Customer Risk Prediction

• Built **LightGBM**-based machine learning model to **predict customer credit risk** from **time-series transaction data**, combining feature engineering, classification modeling, and evaluation techniques to identify high-risk and low-risk segments

• Performed feature engineering on time-series behavioral patterns including temporal statistics, frequency-domain features, and trend indicators, improving model precision by 26% and enhancing interpretability

• Evaluated model performance using precision, recall, and accuracy through reproducible analysis workflows, and designed a decision framework to translate predictions into actionable recommendations